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Curriculum Vitae

Academic appointments

2023-	Professor, Cornell University, Department of Linguistics
2017-	Associate Professor, Cornell University, Department of Linguistics
2017-2018	Visiting Scholar, Wolfson College, Oxford University
2011-2017	Assistant Professor, Cornell University, Department of Linguistics
2009-2011	NIH NRSA Postdoctoral Fellow, University of Southern California Hearing and Communication Neuroscience

Education

2009	Ph.D. Linguistics, University of California, Berkeley Dissertation: Interactions between speech rhythm and gesture
2006	M.A. Linguistics, University of California, Berkeley
2001	B.A. Linguistics, B.A. English, Northwestern University

Fellowships, awards, grants

2022	(Co-PI) Cornell University New Frontiers Grant: Emergence of dialects in networks of speakers with random, constrainted interactions ($198,558)
2016	(Co-PI) Mario Einaudi Center for International Studies Small Grant: LabPhon15: Speech Dynamics and Phonological Representation ($5,000).
2016	(Co-PI) Cornell University Institute for the Social Sciences Small Grant: LabPhon15 - Speech Dynamics and Phonological Representation ($3,500).
2016	(Co-PI) NSF Conference Grant: LabPhon 15 - Speech Dynamics and Phonological Representation, Award #1551588, ($24,845).
2013	Cornell Center for Teaching Excellence Faculty Grant: Earmaster: a phonetic transcription game.
2012	Cornell University Institute for the Social Sciences Small Grant: Visualizing speech: real-time MRI of the vocal tract ($10,680).
2009	NIH Ruth L. Kirschstein-NRSA Institutional Research Training Grant, University of Southern California.
2008	NSF Doctoral Dissertation Improvement Grant, Award #0817767.
2001	Daniel Bonbright Award, Weinberg College of Arts and Sciences, Northwestern University.

Books

2019	**Syntax with oscillators and energy levels**. Tilsen, S. *Language Science Press*.

Manuscripts under review

2026	**A mode-coupling theory of phonological organization**. Tilsen, S. Preprint.

2026 **Ambisyllabicity as decoupling of constriction formation and release**. Tilsen, S. Under review.

Peer-reviewed publications

2026 **Unpredictable temporal feedback perturbation dissociates mismatch detection from correction**. Zhao, F. & Tilsen, S. *Journal of Phonetics*, 118: 101519. doi: 10.1016/j.wocn.2026.101519.

2026 **Mapping the good voice: linguistic applications to the future of voice coaching for performance**. Shipley, M., Reade, A., Tilsen, S., & Babel, M. *Voice and Speech Review*, pp. 1-17. doi: 10.1080/23268263.2026.2720698.

2026 **Phrasal rhythmicity and the sources of temporal intermittency in speech**. Tilsen, S. *Rhythms of Speech and Language*, pp. 274-295. Chapter 20. Cambridge University Press. doi: 10.1017/9781009295888.020.

2025 **The Gesture-Field-Register (GFR) framework for modeling F0 control**. Kim, S.-E. & Tilsen, S. *Journal of Phonetics*, 110: 101410. doi: 10.1016/j.wocn.2025.101410.

2025 **Thai speakers time lexical tones to supralaryngeal articulatory events**. Burroni, F. & Tilsen, S. *Journal of Phonetics*, 108: 101389. doi: 10.1016/j.wocn.2024.101389.

2024 **Planning for the future and reacting to the present: Proactive and reactive F0 adjustments in speech**. Kim, S.-E. & Tilsen, S. *Journal of Phonetics*, 104: 101322. doi: 10.1016/j.wocn.2024.101322.

2024 **Internal speech is faster than external speech: evidence for feedback-based temporal control**. Tilsen, S. *Cognition*, 244: 105713. doi: 10.1016/j.cognition.2023.105713.

2023 **Probability and randomness in phonology: Deep vs. shallow stochasticity**. Tilsen, S. *Studies in Phonetics, Phonology, and Morphology*. doi: 10.17959/sppm.2023.29.2.247.

2023 **Looking within events: examining internal temporal structure with local relative rate**. Tilsen, S. & Tiede, M. *Journal of Phonetics*, 100: 101264. doi: 10.1016/j.wocn.2023.101264.

2023 **Parameters of unit-based measures of speech rate**. Tilsen, S. & Tiede, M. *Speech Communication*, 150: 73-97. doi: 10.1016/j.specom.2023.05.006.

2023 **Hierarchical distinctions in the production and perception of nuclear tunes in American English**. Cole, J., Steffman, J., Shattuck-Hufnagel, S., & Tilsen, S. *Laboratory Phonology*, 14(1). doi: 10.16995/labphon.9437.

2023 **A preliminary study of speech rhythm differences as markers of stuttering persistence in preschool-age children**. Erdemir, A., Walden, T., Mefferd, A., Tilsen, S., & Jones, R. M. *Journal of Speech, Language, and Hearing Research*, 66(3): 931-950. doi: 10.1044/2022_JSLHR-22-00126.

2022 **An informal logic of feedback-based temporal control**. Tilsen, S. *Frontiers in Human Neuroscience*, 16: 851991. doi: 10.3389/fnhum.2022.851991.

2022 **Relaxation, Percolation, and Non-Spontaneous Fluctuation of Linguistic Behavior in a Quasi-Isolated System**. Tilsen, S. *Frontiers in Physics, Social Physics*, 10: 801740. doi: 10.3389/fphy.2022.801740.

2022 **An investigation of functional relations between speech rate and phonetic variables**. Kim, S.-E. & Tilsen, S. *Journal of Phonetics*, 93: 101152. doi: 10.1016/j.wocn.2022.101152.

2022 **The online effect of clash is durational lengthening, not prominence shift: Evidence from Italian**. Burroni, F. & Tilsen, S. *Journal of Phonetics*, 91: 101124. doi: 10.1016/j.wocn.2021.101124.

2021 **Localizing category-related information in speech with multi-scale analyses**. Tilsen, S., Kim, S.-E., & Wang, C. *PLoS ONE*, 16(1): e0258178. doi: 10.1371/journal.pone.0258178.

2021 **Temporal differences between high vowels and glides are more robust than spatial differences**. Burgdorf, D. & Tilsen, S. *Journal of Phonetics*, 88: 101073. doi: 10.1016/j.wocn.2021.101073.

2020 **Detecting anticipatory information in speech with signal chopping**. Tilsen, S. *Journal of Phonetics*, 82: 100996. doi: 10.1016/j.wocn.2020.100996.

2020 **Incongruencies between phonological theory and phonetic measurement**. Mücke, D., Hermes, A., & Tilsen, S. *Phonology*, 37(1): 133-170. doi: 10.1017/S0952675720000068.

2019 **Motoric mechanisms for the emergence of non-local phonological patterns**. Tilsen, S. *Frontiers in Psychology*, 10: 2143. doi: 10.3389/fpsyg.2019.02143.

2019 **Space and time in models of speech rhythm**. Tilsen, S. *Annals of The New York Academy of Sciences*, 1453(1): 47-66. doi: 10.1111/nyas.14102.

2018 **Analysis of Speech Production Real-time MRI**. Ramanarayanan, V., Tilsen, S., Proctor, M., Toger, J., Goldstein, L., Nayak, K., & Narayanan, S. *Computer Speech & Language*, 52: 1-22. doi: 10.1016/j.csl.2018.04.002.

2017 **Exertive modulation of speech and articulatory phasing**. Tilsen, S. *Journal of Phonetics*, 64: 34-50. doi: 10.1016/j.wocn.2017.03.001.

2016 **Shared representations underlie metaphonological judgments and speech motor control**. Tilsen, S. & Cohn, A. *Laboratory Phonology: Journal of the Association for Laboratory Phonology*, 7(1): 14. doi: 10.5334/labphon.52.

2016 **Selection and coordination: the articulatory basis for the emergence of phonological structure**. Tilsen, S. *Journal of Phonetics*, 55: 53-77. doi: 10.1016/j.wocn.2015.11.005.

2016 **Anticipatory Posturing of the Vocal Tract Reveals Dissociation of Speech Movement Plans from Linguistic Units**. Tilsen, S., Spincemaille, P., Xu, B., Doerschuk, P., Luh, W.-M., Feldman, E., & Wang, Y. *PLoS ONE*, 11(1): e0146813. doi: 10.1371/journal.pone.0146813.

2015 **Real-time articulatory biofeedback with electromagnetic articulography**. Tilsen, S., Das, D., & McKee, B. *Linguistics Vanguard*, 1(1): 39-55. doi: 10.1515/lingvan-2014-1006.

2014 **Selection and coordination of articulatory gestures in temporally constrained production**. Tilsen, S. *Journal of Phonetics*, 44: 26-46. doi: 10.1016/j.wocn.2013.12.004.

2013 **A dynamical model of hierarchical selection and coordination in speech planning**. Tilsen, S. *PLoS ONE*, 8(4): e62800. doi: 10.1371/journal.pone.0062800.

2013 **Speech rhythm analysis with decomposition of the amplitude envelope: Characterizing rhythmic patterns within and across languages**. Tilsen, S. & Arvaniti, A. *Journal of the Acoustical Society of America*, 134(1): 628-639. doi: 10.1121/1.4807565.

2013 **Inhibitory mechanisms in speech planning maintain and maximize contrast**. Tilsen, S. In A. Yu (Ed.), *Origins of Sound Patterns: Approaches to Phonologization*, pp. 112-127. Oxford: Oxford University Press.

2012 **Articulatory gestures are individually selected in production**. Tilsen, S. & Goldstein, L. *Journal of Phonetics*, 40: 764-779.

2012 **Utterance preparation and stress clash: planning prosodic alternations**. Tilsen, S. In S. Fuchs, P. Perrier, M. Weirich, & D. Pape (Eds.), *Speech production and perception: Planning and dynamics*, pp. 119-156. Frankfurt am Main: Peter Lang Verlag.

2011 **Effects of syllable stress on articulatory planning observed in a stop-signal experiment**. Tilsen, S. *Journal of Phonetics*, 39: 642-659.

2011 **Metrical regularity facilitates speech planning and production**. Tilsen, S. *Laboratory Phonology*, 2(1): 185-218.

2010 **Automated measurement of vowel formants in the Buckeye Corpus**. Yao, Y., Tilsen, S., Sprouse, R., & Johnson, K. *Gengo Kenkyu (Journal of the Linguistic Society of Japan)*, 138: 99-113.

2009 **Multi-timescale dynamical interactions between speech rhythm and gesture**. Tilsen, S. *Cognitive Science*, 33: 839-879.

2009 **Subphonemic and cross-phonemic priming in vowel shadowing: evidence for the involvement of exemplars in production**. Tilsen, S. *Journal of Phonetics*, 37(3): 276-296.

2008 **Low-frequency Fourier analysis of speech rhythm**. Tilsen, S. & Johnson, K. *Journal of the Acoustical Society of America*, 124(2): EL34-39.

Conference proceedings

2025 **Envelope-based metrics reveal speech rhythm development tendencies in Brazilian L2 English learners**. Teixeira, L. A. S., Tilsen, S., Lima, R. M., & Arantes, P. Preprint.

2023 **EchoNose: Sensing Mouth, Breathing and Tongue Gestures inside Oral Cavity using a Non-contact Nose Interface**. Sun, R., Zhou, X., Steeper, B., Zhang, R., Yin, S., Li, K., Wu, S., Tilsen, S., Guimbretiere, F., & Zhang, C. *Proceedings of the 2023 International Symposium on Wearable Computers*, pp. 22-26. Association for Computing Machinery. doi: 10.1145/3594738.3611358.

2023 **Morpho-phonological effects on the phonetic characteristics of tense consonants in Korean compounds**. Kwon, C. D., Tilsen, S., & Whitman, J. *Proc. of the 20th International Congress of Phonetic Sciences*.

2023 **Investigations of F0 control: pitch targets vs pitch register**. Kim, S.-E. & Tilsen, S. *Proc. of the 20th International Congress of Phonetic Sciences*.

2022 **Shape matters: Machine classification and listeners' perceptual discrimination of American English intonational tunes**. Cole, J., Tilsen, S., & Steffman, J. *Proc. Speech Prosody 2022*, pp. 297-301.

2021 **Articulatory characteristics of Icelandic voiced fricative lenition: gradience, categoricity, and speaker/gesture-specific effects**. Stefansdottir, B., Burroni, F., & Tilsen, S. *Proc. Interspeech 2021*.

2021 **Compensation for altered feedback in vowels and glides**. Burgdorf, D. & Tilsen, S. *Proc. of the 12th International Seminar on Speech Production*.

2021 **Temporal localization of syntactically conditioned prosodic information**. Kim, S.-E. & Tilsen, S. *Proc. of the 12th International Seminar on Speech Production*.

2021 **Nonlinear effects of speech rate on articulatory timing in singletons and geminates**. Tilsen, S. & Hermes, A. *Proc. of the 12th International Seminar on Speech Production*, pp. 56-59.

2021 **Cross-linguistic timing contrast in geminates: a rate-independent perspective**. Hermes, A., Tilsen, S., & Ridouane, R. *Proc. of the 12th International Seminar on Speech Production*, pp. 52-55.

2020 **Prominence clash does not induce rhythmic adjustments in Italian**. Burroni, F. & Tilsen, S. *Proc. Speech Prosody 2020*, pp. 571-575. doi: 10.21437/SpeechProsody.2020-117.

2020 **Speech rate and syntactically conditioned influences on prosodic boundaries**. Kim, S.-E. & Tilsen, S. *Proc. Speech Prosody 2020*, pp. 434-438. doi: 10.21437/SpeechProsody.2020-89.

2019 **Strength and structure: coupling tones with oral constriction gestures**. Mücke, D., Hermes, A., & Tilsen, S. *Proc. Interspeech 2019*, pp. 914-918. doi: 10.21437/Interspeech.2019-2650.

2016 **A shared control parameter for F0 and intensity**. Tilsen, S. *Proc. Speech Prosody 2016*, pp. 1066-1070.

2016 **Early Prosodic Manifestations of Disfluency**. Li, J. & Tilsen, S. *Proc. Speech Prosody 2016*, pp. 1235-1239.

2015 **Structured nonstationarity in articulatory timing**. Tilsen, S. *Proc. of the 18th International Congress of Phonetic Sciences*.

2015 **Relation between syllable count judgments and durations of English liquid rimes**. Cohn, A. & Tilsen, S. *Proc. of the 18th International Congress of Phonetic Sciences*.

2015 **Phonetic evidence for two types of disfluency**. Li, J. & Tilsen, S. *Proc. of the 18th International Congress of Phonetic Sciences*.

2015 **The articulatory tone-bearing unit: Gestural coordination of lexical tone in Thai**. Karlin, R. & Tilsen, S. *POMA*, 22: 060006. doi: 10.1121/2.0000089.

2015 **Gestural timing in Mandarin tone sandhi**. Yi, H. & Tilsen, S. *POMA*, 22: 060003. doi: 10.1121/2.0000029.

2015 **A real-time MRI investigation of anticipatory posturing in prepared responses**. Tilsen, S., Spincemaille, P., Xu, B., Doerschuk, P., Karlin, R., Yi, H., Feldman, E., Luh, W., & Wang, Y. *POMA*, 22: 060001. doi: 10.1121/2.0000006.

2014 **Some people are eager to be heard: anticipatory posturing in speech production**. Tilsen, S., Spincemaille, P., Xu, B., Doerschuk, P., Karlin, R., Yi, H., Feldman, E., Luh, W., & Wang, Y. *168th Meeting of the Acoustical Society of America Lay Language Papers*.

2013 **Comparison of native and non-native consonant articulation with real-time magnetic resonance imaging of the vocal tract**. Tilsen, S., Xu, B., Spincemaille, P., Srivastava, M., Doerschuk, P., & Wang, Y. *POMA*, 19: 060246.

2013 **Acoustic and articulatory evidence for the phonological status of liaison consonants**. L'Esperance, M.-J. & Tilsen, S. *POMA*, 19: 060245.

2008 **Relations between speech rhythm and segmental deletion**. Tilsen, S. *Proc. of the Annual Meeting of the Chicago Linguistic Society*, 44(1): 221-223.

Working papers

2022 **Interpreting phonetic evidence for hierarchical organization of prosodic phrases**. Kim, S.-E. & Tilsen, S. *Cornell Working Papers in Phonetics and Phonology*. doi: 10.13140/RG.2.2.14146.61120.

2020 **Exploratory investigations of SVO sentence production: evidence for category-specific interference effects**. Tilsen, S. *Cornell Working Papers in Phonetics and Phonology*.

2020 **A different view of gestural activation: learning gestural parameters and activation with an RNN**. Tilsen, S. *Cornell Working Papers in Phonetics and Phonology*.

2018 **Three mechanisms for modeling articulation: selection, coordination, and intention**. Tilsen, S. *Cornell Working Papers in Phonetics and Phonology*.

2016 **Correlations between speech behavior and social network structure in a constrained vocabulary game**. Tilsen, S. *Cornell Working Papers in Phonetics and Phonology*.

2015 **Speech and social network dynamics in a constrained vocabulary game: design and hypotheses**. Tilsen, S. *Cornell Working Papers in Phonetics and Phonology*.

2014 **Syllable count judgments and durations of liquid rimes in English**. Tilsen, S., Cohn, A., & Ricciardi, E. *Cornell Working Papers in Phonetics and Phonology*. doi: 10.5281/zenodo.3726289.

2014 **Selection-coordination theory**. Tilsen, S. *Cornell Working Papers in Phonetics and Phonology*. doi: 10.5281/zenodo.3726282.

2013 **Imitation of intonational gestures: a preliminary report**. Tilsen, S., Burgess, D., & Lantz, E. *Cornell Working Papers in Phonetics and Phonology*. doi: 10.5281/zenodo.3726927.

2012 **Acoustic analysis of the effects of metrical regularity on interval durations**. Tilsen, S., Cibelli, E., Kang, S., Houde, J., & Nagarajan, S. *Cornell Working Papers in Phonetics and Phonology*. doi: 10.5281/zenodo.3726943.

2012 **A cross-linguistic investigation of articulatory coordination in word-initial consonant clusters**. Tilsen, S., Zec, D., Bjorndahl, C., Butler, B., L'Esperance, M. J., Fisher, A., Heimisdottir, L., Renwick, M., & Sanker, C. *Cornell Working Papers in Phonetics and Phonology*. doi: 10.5281/zenodo.3726936.

2009 **Toward a dynamical interpretation of hierarchical linguistic structure**. Tilsen, S. *UC Berkeley Phonology Lab Annual Report*, pp. 462-512.

2008 **Preliminary results of a stop-signal experiment**. Tilsen, S. *UC Berkeley Phonology Lab Annual Report*, pp. 250-268.

2007 **Vowel-to-vowel coarticulation and dissimilation in response-priming**. Tilsen, S. *UC Berkeley Phonology Lab Annual Report*, pp. 416-458.

2007 **Low-frequency Fourier analysis of speech rhythm**. Tilsen, S. & Johnson, K. *UC Berkeley Phonology Lab Annual Report*, pp. 686-712.

2006 **Rhythmic coordination in repetition disfluency: a harmonic timing effect**. Tilsen, S. *UC Berkeley Phonology Lab Annual Report*, pp. 73-114.